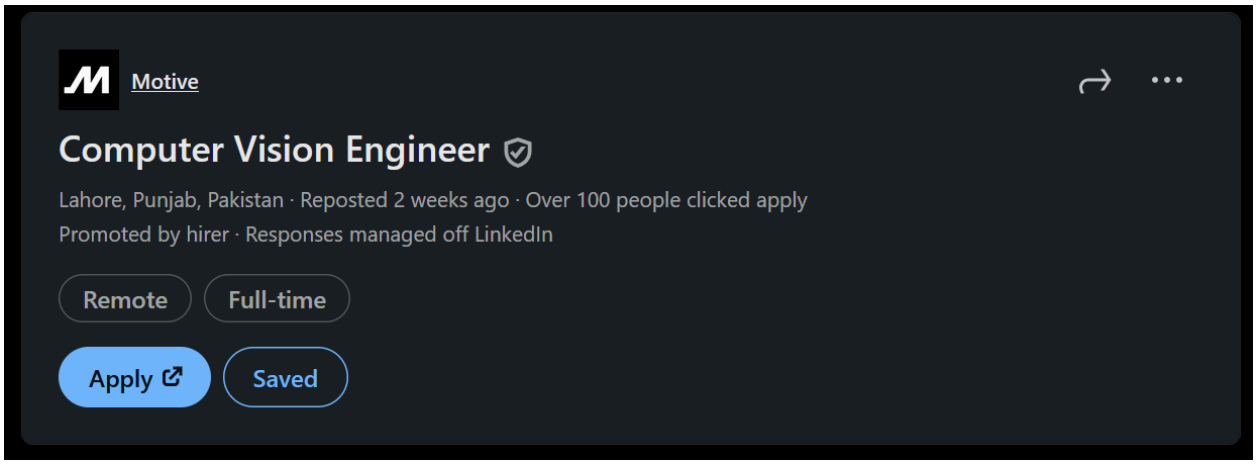
Assignment 3

Job Listing:



About The Role

As a Computer Vision Engineer, you will be a part of a passionate team whose mission is to bring intelligence to the world's trucks. The team is focused on building technology to understand driving behavior, identify risk factors, and intelligently suggest coachable events that not only improves the fleet safety and potentially save millions of dollars but also contribute to making the roads safer. You will have a unique opportunity to work with a high caliber and fast-paced team which consists of experienced researchers and engineers in Computer Vision, Machine Learning, Deep Learning, and Robotics with a track record of previous products and top-tier publications.

You will play a critical role in building and improving a technology that will be used by millions of trucks. In this role, you will design and implement complex computer vision systems. You will have the opportunity to build and/or improve algorithms for edge processing, implement cloud services, and integrate them to deliver end-to-end product features. You will also get exposure to large-scale ML infra and scaling that facilitates large amounts of data to train, test, and validate computer vision systems.

What You'll Do

- Improve the performance of our models and algorithms already in production
- Build and optimize CV/ML algorithms for real-time performance so they can run on our embedded platform i.e. the next-gen AI dashcam
- Write proficient Python and C++ code to build and improve CV algorithms, ML services, training, model compression and porting pipelines
- Convert state-of-the-art research prototypes into reliably working product features
- Collaborate with RnD team working in Computer Vision, Machine Learning, and Robotics, along with ML Platform, Embedded, Backend, and Frontend teams to deliver end-to-end product features
- Solve challenging object detection, segmentation, recognition, and geometric computer vision problems.
- Build automated deployment, validation, and active learning pipelines.

What We're Looking For

- Bachelors degree in computer science, software engineering, data science, artificial intelligence, or related field.
- Masters or PhD in Computer Vision, Deep Learning, and Machine Learning (preferred)
- Solid mathematical foundation in Computer Vision, Deep Learning, Machine Learning, and optimization approaches.
- Hands-on experience in training framework and tools (TensorFlow | Pytorch | Caffe)
- Strong experience in Python or C++
- Hands-on experience in building embedded algorithms and applications
- Experience in the following tools and technologies is a plus: AWS (SageMaker, Lambda, EC2, S3, RDS), CI/CD, Terraform, Docker, and Kubernetes.
- Prior experience with Eigen library and RTOS is a strong plus

Can be viewed on LinkedIn at: https://www.linkedin.com/jobs/view/4044252568

Cover Email:

Subject: Application for Computer Vision Engineer Position @ Motive

Dear Hiring Manager,

I am writing to express my interest in the CV Engineer position recently advertised by your organization. As a Data Science undergraduate at FAST-NUCES with hands-on experience in machine learning, data visualization, and model deployment, I am eager to contribute to your team's data-driven initiatives.

During my internship at atomcamp, I developed and optimized forecasting and computer vision models, achieving up to 93% prediction accuracy and a 30% improvement in system efficiency. I also engineered real-time facial recognition pipelines using InsightFace and YOLOv8, later deploying them through AWS and Docker. These experiences enhanced my technical proficiency in Python, SQL, and TensorFlow while strengthening my ability to manage projects under tight deadlines.

I am particularly drawn to your company's focus on innovation and applied AI solutions. My goal is to contribute to data-centric problem solving and scalable model design while further developing my analytical and engineering skills in a professional setting.

Thank you for considering my application. I have attached my resume for your review and would appreciate the opportunity to discuss how I can contribute to your team.

Sincerely,

Muhammad Muneeb Lone

muneeblone0000@gmail.com

Resume:

Below is an image of my resume. It is also attached as a pdf alongside this pdf on Google Classroom.

Muhammad Muneeb Lone

Rawalpindi | muneeblone0000@gmail.com | linkedin.com/in/muneeb-lone | github.com/Mnb-0

Objective

Detail-oriented and mathematically grounded Data Science undergraduate seeking to apply hands-on experience in deep learning, computer vision, and embedded systems to develop scalable, real-time perception solutions.

Education

FAST-NUCES, BSc Data Science

- GPA: 3.3/4.0
- **Relevant Coursework:** Multivariable Calculus, Linear Algebra, Probability and Statistics, Advanced Statistics, Introduction To Data Science, Object Oriented Programming, Data Structures, Database Systems

Experience

AI Engineer (Freelance), atomcamp – Islamabad

- Designed and deployed a facial recognition pipeline using InsightFace, achieving 97% accuracy on real-world video data.
- Reduced model inference latency by 22% through quantization and multithreading, enabling real-time performance on embedded systems.
- Deployed scalable CV systems via Docker and AWS EC2; monitored and optimized workloads to lower runtime costs by 18%.

Machine Learning Intern, atomcamp – Islamabad

- Developed and evaluated forecasting models (LightGBM, Prophet, NeuralProphet, LSTM) with up to 93% prediction accuracy.
- Refactored and modularized a YOLOv8-InsightFace surveillance system, improving processing efficiency by 30%.
- Integrated AWS deployment pipelines, SQLite-based logging, and image capture modules to streamline data operations.

Projects

Forecasting Inflation in Pakistan

- Modeled inflation trends (1973–2023) using ARIMA and regularized regression (Ridge, LASSO, Elastic Net).
- Processed 10,000+ data points and improved prediction accuracy by 15% after cross-validation.
- Tools Used: R, tidyverse, GGally, forecast, glmnet, ggplot2

Predictive Modeling and Customer Clustering for Retail Optimization

- Performed data cleaning, exploratory analysis, predictive modeling, and segmentation on 50,000+ retail records.
- Identified key customer segments with K-Means, leading to a simulated 12% improvement in retention rate.
- Tools Used: Python, Pandas, Scikit-Learn, Seaborn, Statsmodels

Technologies

Languages: C, C++, C#, Python, R, SQL, x86 Assembly (MASM), Bash

Libraries & Frameworks: NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, NLTK, OpenCV, TensorFlow/Keras/PyTorch, Optuna

Technologies & Tools: Linux, WSL2, Git, GitHub, MySQL, Kaggle, Google Colab, AWS

Achievements

- Dean's List – FAST-NUCES
- High Achiever's Award – A Levels
- High Achiever's Award – O Levels
- Head of Operations – FAST Data Science Society